CLASS 9 · CBSE · GANITA MANJARI PART 1 · FORMULA BANK

Chapter 3: The World of Numbers

Every formula and result of the chapter on one sheet

1	Rational number	$\frac{p}{q}$, with p, q integers and $q \neq 0$
2	Equality	$\frac{a}{b} = \frac{c}{d}$ when $ad = bc$
3	Product	$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$
4	Quotient	$\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$
5	Distributive law	$p(q + r) = pq + pr$
6	Rules for zero	$a + 0 = a$, $a - 0 = a$, $a \times 0 = 0$
7	Signs in multiplication	$(-) \times (+) = (-)$ and $(-) \times (-) = (+)$
8	Absolute value	$\lvert x \rvert$ = distance of x from 0; $\lvert x \rvert \geq 0$
9	Distance between a and b	$\lvert a - b \rvert$
10	A rational number between a and b	$\frac{a + b}{2}$
11	Terminating decimal	In lowest form, the denominator has no prime factor other than 2 and 5
12	Number of decimal places	For $q = 2^m \times 5^n$, the larger of m and n
13	Pure repeating decimal	Multiply by 10^n (n repeating digits) and subtract
14	General repeating decimal	Multiply by 10^m, then by 10^n, and subtract
15	Diagonal of a unit square	$\sqrt{2}$